

Reinforcement Learning Agent for Optimization of an H₂S Stripper Column

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Abstract—This paper presents a reinforcement learning (RL) agent for optimizing the recovery of H₂S/NH₃ on an sour water stripper column without direct concentration measurements. Two actor–critic algorithms, Proximal Policy Optimization (PPO) and Deep Deterministic Policy Gradient (DDPG), are trained inside a deep neural network (DNN) surrogate environment built from Aspen-simulated data sampled via Latin Hypercube Sampling (LHS). Hyperparameters and reward weights are jointly optimized using Optuna Bayesian optimization (50 trials each). The formulation is cast as a Partially Observable Markov Decision Process (POMDP), making feed concentration and Murphree efficiency latent variables unknown to the agent. Validation against a static Aspen simulation shows that both agents achieves H₂S recoveries above xxxx. PPO reaches a mean H₂S recovery of 99.98 % with xxx Aspen convergence iterations per episode, whereas DDPG-Opt achieves 99.86 % at xxx iterations. Results confirm that surrogate-trained RL with Optuna tuning is a practical and computationally efficient approach for industrial process optimization without on-line composition analyzers.

Index Terms—Reinforcement learning, process optimization, surrogate model, Sour Water, H₂S stripping, Optuna, PPO, DDPG, POMDP.

I. INTRODUCTION

Sour-water stripper columns in petroleum refining remove hydrogen sulfide (H₂S) and ammonia (NH₃) from process water, generating overhead streams suitable for recovery or thermal destruction in sulfur-recovery units. Efficient operation requires maintaining the reboiler duty $\dot{Q}_{reb}$ near the so-called *operational knee-point*: the region where incremental energy input yields diminishing returns in stripping efficiency. Operating below this point leaves contaminants in the stripped water; operating above it wastes energy and may violate hydraulic constraints.

On-line H₂S and NH₃ analyzers are expensive, require intensive maintenance, and are often unavailable in practice. Operators therefore rely on indirect temperature indicators—bottom temperature T_{bottom} and top temperature T_{top} —to infer column performance, adjusting $\dot{Q}_{reb}$ empirically. This manual, reactive strategy rarely achieves the optimum and is sensitive to unmeasured disturbances such as changes in feed loading, column pressure, or tray efficiency.

Reinforcement learning has emerged as a promising approach for process optimization, particularly when the optimal operating policy is difficult to express analytically [1]–[3]. RL agents learn by interacting with a simulated environment and can naturally handle nonlinear, time-varying processes. A key practical barrier, however, is the computational cost of

process simulation; rigorous simulators may require minutes per steady-state evaluation, making the millions of interactions typical of RL training prohibitive.

Surrogate models—fast approximations of high-fidelity simulators—address this bottleneck [4], [5]. Deep neural network (DNN) surrogates trained on simulation data can be queried in microseconds, enabling intensive RL training at negligible computational cost. The policy can then be validated against the rigorous simulator to quantify model bias.

This work makes the following contributions:

- A POMDP formulation of the knee-point optimization problem in which feed concentrations and Murphree tray efficiency are treated as latent (unmeasured) variables, forcing the agent to learn a robust policy;
- A two-model DNN surrogate for steady-state prediction of T_{bottom} and T_{top} , trained on 700 LHS-sampled Aspen simulations;
- Joint Bayesian optimization of RL hyperparameters and reward weights using Optuna, applied to both PPO and DDPG;
- Static Aspen validation of the trained policies, reporting H₂S recovery, NH₃ loss, and convergence behavior across 30 episodes per agent.

II. PROCESS DESCRIPTION

The system studied is a single-feed, single-reboiler sour-water stripper operating at approximately 6.0 kgf/cm²g. The column receives a sour-water stream loaded with H₂S (875–3500 ppm) and NH₃ (750–3000 ppm) at variable flow rates (47,500–52,500 kg/h) and temperatures (119–125 °C). Steam is supplied to the reboiler, which manipulates $\dot{Q}_{reb}$ to drive stripping. The overhead vapor is sent to downstream treatment; the bottoms product must meet H₂S and NH₃ discharge specifications.

The steady-state behavior of the column is described by the MESH equations implemented in Aspen Plus with Murphree tray efficiency. Tray efficiency E_{column} is a global scalar applied uniformly to all trays; its value is uncertain and may drift over time due to fouling or changes in liquid distribution. Column pressure P_{column} also varies within its normal operating band. These two parameters constitute the main sources of unmeasured process variability.

The operational knee-point manifests as a change in the slope of the T_{bottom} -vs- $\dot{Q}_{reb}$ curve: below the knee, additional heat input produces a significant temperature increase indicating improved stripping; above it, the curve flattens as the

column approaches thermodynamic saturation and energy is dissipated without proportional stripping benefit.

III. METHODOLOGY

A. DNN Surrogate Model

1) *Dataset Generation via LHS*: A space-filling Latin Hypercube Sampling (LHS) design with a maximin criterion was used to generate 1000 candidate operating points in the seven-dimensional input space

$$\mathbf{x} = [Q_{\text{feed}}, T_{\text{feed}}, c_{\text{H}_2\text{S}}, c_{\text{NH}_3}, \dot{Q}_{\text{reb}}, P_{\text{column}}, E_{\text{column}}].$$

Each point was simulated in Aspen Plus to steady-state convergence. Points that did not converge or violated hydraulic constraints were flagged as invalid (`valid = 0`) and excluded from regression training. The final usable dataset comprised 1000 samples, split into 700 training, 150 validation, and 150 test samples.

2) *Architecture and Training*: Two independent feedforward DNN regressors were trained—one for T_{bottom} and one for T_{top} —using Keras with TensorFlow. Hyperparameters (layer width, depth, learning rate, dropout, L2 regularization, batch size) were tuned via a separate Optuna study for each surrogate. All seven inputs were normalized to $[0, 1]$ using the min-max ranges of the dataset; outputs were normalized in the same manner and then denormalized during inference.

Final surrogate architectures and performance metrics are reported in Table I.

Table I
SURROGATE DNN PERFORMANCE ON THE HELD-OUT TEST SET.

Metric	T_{bottom} model	T_{top} model
Hidden layers	1 (224 neurons)	2 (160+192 neurons)
Learning rate	2.86×10^{-3}	9.24×10^{-3}
Batch size	16	16
Dropout rate	0.025	0.003
Training epochs	226	253
RMSE (°C)	0.0063	0.0111
R^2 (validation)	0.9992	0.9995
R^2 (test)	0.9992	0.9998

Both surrogates achieve $R^2 > 0.999$ on the test set, with RMSE below 0.012 °C, indicating accuracy well within the tolerance required for RL training.

B. Reinforcement Learning Formulation

1) *POMDP Formulation*: The optimization problem is formulated as a Partially Observable Markov Decision Process (POMDP), since feed concentrations ($c_{\text{H}_2\text{S}}$, c_{NH_3}) and Murphree efficiency (E_{column}) affect the environment but are unavailable to the agent. The true environment state is

$$\mathbf{x}_t = [Q_{\text{feed}}, T_{\text{feed}}, c_{\text{H}_2\text{S}}, c_{\text{NH}_3}, \dot{Q}_{\text{reb}}, P_{\text{column}}, E_{\text{column}}],$$

while the agent observation is restricted to the six measurable variables:

$$\mathbf{s}_t = [Q_{\text{feed}}, T_{\text{feed}}, \dot{Q}_{\text{reb}}, P_{\text{column}}, T_{\text{bottom}}, T_{\text{top}}].$$

2) *Action Space*: The action is a continuous, incremental adjustment to the reboiler duty setpoint:

$$a_t \in [-1, +1] \Rightarrow \Delta \dot{Q}_{\text{reb}} = a_t \times 0.3 \text{ Gcal/h}, \quad (1)$$

with the updated setpoint clipped to the safe operating range:

$$\dot{Q}_{\text{reb}}(t+1) = \text{clip}(\dot{Q}_{\text{reb}}(t) + \Delta \dot{Q}_{\text{reb}}, 2.4, 6.0) \text{ Gcal/h}. \quad (2)$$

The incremental formulation avoids abrupt setpoint changes, guarantees smooth trajectories, and allows the agent to traverse the full operating range in approximately 12 steps within a 30-step episode.

3) *Reward Function*: A single reward formulation (Reward A) was adopted, combining three operational objectives:

$$r_t = w_1 \cdot (\tilde{T}_{\text{bottom}} - \tilde{T}_{\text{top}}) - w_2 \cdot \tilde{\dot{Q}}_{\text{reb}} - w_3 \cdot \text{Var}(T_{\text{bottom}}), \quad (3)$$

where tilde notation denotes min-max normalization over the operating range, and $\text{Var}(T_{\text{bottom}})$ is computed over a sliding window to penalize oscillation. A safety penalty of -50 is applied whenever T_{bottom} or T_{top} exceed their operating limits, and the episode is terminated. The weights (w_1, w_2, w_3) are treated as hyperparameters and jointly optimized by Optuna.

4) *Episode Initialization and Domain Randomization*: Each training episode randomizes:

- *Observed disturbances*: Q_{feed} and T_{feed} drawn uniformly from their operating ranges;
- *Latent variables*: $c_{\text{H}_2\text{S}}$, c_{NH_3} , E_{column} drawn uniformly, unseen by the agent;
- *Initial $\dot{Q}_{\text{reb}}$* : sampled from one of three bands (low, mid, high) with equal probability, forcing the agent to learn approach strategies from below and above the optimum.

C. Actor-Critic Algorithms

1) *PPO (Proximal Policy Optimization)*: PPO [3] is an on-policy actor-critic algorithm that constrains policy updates via a clipped surrogate objective, providing stable learning without requiring precise step-size tuning. It is particularly well-suited to environments where sample efficiency is less critical than training stability.

2) *DDPG (Deep Deterministic Policy Gradient)*: DDPG [2] is an off-policy algorithm that maintains a replay buffer and uses delayed target networks. Exploration is driven by Ornstein-Uhlenbeck noise added to the deterministic policy output. The off-policy nature allows higher sample efficiency but requires careful tuning of the noise level and learning rates.

Both algorithms were implemented using the `stable-baselines3` library [7].

D. Hyperparameter Optimization with Optuna

Optuna [6] was used to jointly optimize RL hyperparameters and reward weights through Tree-structured Parzen Estimator (TPE) sampling. The optimization protocol comprised two phases:

- 1) **Search phase**: 50 independent trials, each training the agent for 150,000 timesteps; the objective is the mean episodic reward over the final evaluation episodes.

- 2) **Retraining phase:** the best-performing hyperparameter configuration is retrained from scratch for 500,000 timesteps.

The search spaces are summarized in Tables II and III.

Table II
OPTUNA SEARCH SPACE FOR PPO.

Parameter	Type	Range
learning_rate	log-float	$[10^{-5}, 10^{-3}]$
n_steps	categorical	512, 1024, 2048, 4096
batch_size	categorical	32, 64, 128, 256
n_epochs	int	[4, 20]
gamma	float	[0.95, 0.999]
gae_lambda	float	[0.90, 1.00]
clip_range	float	[0.10, 0.40]
ent_coef	float	[0.00, 0.10]
w_1	float	[0.05, 0.30]
w_2	float	[0.50, 1.50]
w_3	float	[0.10, 1.50]

Table III
OPTUNA SEARCH SPACE FOR DDPG.

Parameter	Type	Range
learning_rate	log-float	$[10^{-4}, 10^{-2}]$
buffer_size	categorical	50k, 100k, 200k
learning_starts	int	[500, 5000]
batch_size	categorical	64, 128, 256, 512
tau	float	[0.001, 0.05]
gamma	float	[0.90, 0.999]
train_freq	categorical	1, 2, 4
gradient_steps	categorical	1, 2, 4
noise σ	float	[0.01, 0.30]
w_1	float	[0.05, 0.30]
w_2	float	[0.50, 1.50]
w_3	float	[0.10, 1.50]

IV. RESULTS

A. Optuna Optimization Results

Table IV reports the best trial found by Optuna for each algorithm and the corresponding optimized hyperparameters.

Table IV
OPTUNA BEST TRIAL RESULTS.

Parameter	PPO-Opt	DDPG-Opt
Best trial	#31 / 50	#46 / 50
Best reward (150k steps)	831.81	783.49
learning_rate	1.897×10^{-4}	4.018×10^{-4}
batch_size	256	256
gamma	0.9712	0.9554
n_steps / buffer_size	512	100k
n_epochs / train_freq	20	4
noise σ	—	0.2407
w_1	0.2997	0.2867
w_2	0.5055	0.6504
w_3	0.6969	0.8614

Both agents selected similar reward weights, with w_1 near its upper bound (maximizing thermal gradient) and moderately weighted energy penalty w_2 . PPO achieved a higher best-trial reward (831.81 vs. 783.49), reflecting faster convergence within the 150k-step budget.

B. Training Performance (500,000 Timesteps)

After retraining with the best hyperparameters for 500,000 timesteps, both agents showed stable policy learning. Table V summarizes key training metrics.

Table V
TRAINING PERFORMANCE OVER 500,000 TIMESTEPS.

Metric	PPO-Opt	DDPG-Opt
Best mean reward	863.03	812.24
Total training episodes	16,674	16,666
Eval return mean	217.30	217.11
Eval return std	41.54	41.43

The nearly identical evaluation return (mean ≈ 217 , std ≈ 41) for both agents suggests that both policies converge to comparable operational setpoints despite different learning dynamics. PPO achieves a higher peak reward (863 vs. 812) due to on-policy updates being better calibrated by the tuned clip range.

C. Aspen Static Validation

After training, each agent was used to simulate 30 episodes (10 per $\dot{Q}_{\text{reb}}$ initialization band: low, mid, high) using the surrogate environment. The final $\dot{Q}_{\text{reb}}$ from each episode was then used as input to Aspen Plus for a static (steady-state) simulation, recovering the actual H_2S and NH_3 concentrations in the bottoms stream.

Full validation results are reported in Table VI.

Table VI
ASPEN STATIC VALIDATION RESULTS (30 EPISODES PER AGENT, MEAN $\pm$ STD).

Metric	PPO-Opt	DDPG-Opt
$\dot{Q}_{\text{reb}}$ final (Gcal/h)	2.679 ± 0.144	2.676 ± 0.134
T_{bottom} ($^{\circ}\text{C}$)	166.05 ± 0.93	166.06 ± 0.96
T_{top} ($^{\circ}\text{C}$)	60.82 ± 0.28	60.70 ± 0.25
CH_2S input (ppm)	2133.68 ± 857.81	2094.84 ± 845.73
CNH_3 input (ppm)	1917.06 ± 650.81	1903.02 ± 657.68
H_2S recovery (%)	99.98 $\pm$ 0.09	99.86 ± 0.68
NH_3 loss (%)	0.72 ± 0.50	0.50 ± 0.43
Aspen iterations (conv.)	5.53 ± 3.69	16.50 ± 12.37

Key observations:

a) *Reboiler duty convergence:* Both agents converge to nearly identical $\dot{Q}_{\text{reb}}$ setpoints (2.679 and 2.676 Gcal/h, respectively), significantly below the upper operating limit of 6.0 Gcal/h. This consistent low-duty setpoint is characteristic of operation at the knee-point: the agents identified that additional energy input beyond ≈ 2.68 Gcal/h does not improve stripping performance proportionally, precisely the defining feature of the knee.

b) *Temperature profiles:* Both agents achieve essentially the same T_{bottom} (≈ 166.05 $^{\circ}\text{C}$) and T_{top} (≈ 60.8 $^{\circ}\text{C}$), confirming convergence to the same thermodynamic operating point independent of algorithm.

c) H_2S recovery: PPO-Opt achieves a mean H_2S recovery of 99.98%, virtually complete removal, with very low variance ($\pm 0.09\%$). DDPG-Opt achieves 99.86% with higher variance ($\pm 0.68\%$), suggesting that PPO's policy is marginally more robust across different feed conditions and initialization bands.

d) NH_3 loss: DDPG-Opt exhibits slightly lower NH_3 loss (0.50% vs. 0.72%), a secondary objective that could be relevant depending on downstream treatment requirements.

e) *Aspen convergence*: A notable difference is in Aspen convergence behavior: PPO-Opt operating points converge in 5.5 iterations on average, while DDPG-Opt requires 16.5 iterations. This suggests that DDPG-Opt's selected setpoints sit closer to regions of slower convergence in the Aspen solver, potentially associated with operating closer to hydraulic boundaries or regions of stronger component interaction.

D. Robustness to Latent Disturbances

Since c_{H_2S} , c_{NH_3} , and E_{column} were randomized per episode but never revealed to the agent, the validation results implicitly test policy robustness. The low standard deviations in $\dot{Q}_{\text{reb}}$ (0.14 for PPO, 0.13 for DDPG) despite wide input concentration ranges (875–3500 ppm H_2S , 750–3000 ppm NH_3) indicate that both agents learned to infer near-optimal setpoints from thermal measurements alone, confirming the POMDP formulation's practical viability.

V. DISCUSSION

A. Comparison with Conventional Approaches

Traditional supervisory optimization for stripper columns typically relies on first-principles models updated via parameter estimation (e.g., online reconciliation of efficiency) or static gain matrices calibrated at a nominal operating point. Both approaches require either periodic re-identification or plant tests, are sensitive to model mismatch, and cannot easily incorporate bounds on unmeasured disturbances. The proposed RL framework requires no model update post-training: robustness to latent variability is achieved at training time through domain randomization.

B. Surrogate Accuracy and Policy Transferability

The DNN surrogates achieve RMSE below 0.012 °C, which is far smaller than the thermal gradient exploited by the reward function ($T_{\text{bottom}} - T_{\text{top}} \approx 105$ °C at the knee). This high accuracy ensures that the policy trained on the surrogate transfers reliably to the Aspen simulator, as confirmed by the validation results. The primary source of residual error is the simplified treatment of tray hydraulics and phase equilibria near column capacity limits.

C. PPO vs. DDPG

Both algorithms converge to essentially the same operating point, but PPO-Opt outperforms DDPG-Opt on H_2S recovery (99.98% vs. 99.86%) and Aspen convergence speed (5.5 vs. 16.5 iterations). PPO's on-policy updates, regularized by the clipped surrogate objective, appear to produce smoother

setpoint trajectories that land in regions where the rigorous simulator converges more easily. DDPG's off-policy learning may introduce subtle distribution shifts that push some operating points toward less numerically stable regions. However, DDPG's marginally lower NH_3 loss suggests a slightly different balance between the two competing objectives in its learned policy.

D. Role of Optuna in Reward Design

Jointly optimizing reward weights w_1 , w_2 , w_3 with RL hyperparameters proved critical: both agents selected w_1 values near the upper limit of the search space, indicating that thermal gradient maximization is the dominant signal for locating the knee-point. The energy penalty weight w_2 was found in the range 0.50–0.65, balancing efficiency without suppressing exploration of higher-duty regions during training. This automated reward calibration avoids the manual reward engineering that often limits RL applicability in industrial settings.

E. Limitations and Future Work

The current validation uses a static Aspen simulation, which captures steady-state behavior but not transient dynamics. A dynamic Aspen Dynamics simulation would be required to validate the incremental control strategy under realistic ramp rates and disturbance rejection scenarios. Additionally, fine-tuning the surrogate using active learning (querying Aspen at states visited by the RL policy) could reduce model bias in the knee-point region. Finally, deploying the agent on a pilot plant or digital twin would allow real-time feedback on closed-loop behavior under actual noise and delay conditions.

VI. CONCLUSION

This work demonstrated that a surrogate-trained RL supervisory agent can reliably locate and maintain the operational knee-point of an H_2S/NH_3 stripper column using only thermal and flow measurements—without on-line composition analyzers. The POMDP formulation, combined with domain randomization over feed concentrations and Murphree efficiency, yielded policies robust to unmeasured process variability.

Two actor-critic algorithms (PPO and DDPG) were evaluated, both augmented with Optuna Bayesian hyperparameter and reward-weight optimization. Static Aspen validation over 30 episodes per agent showed that:

- Both agents converge to $\dot{Q}_{\text{reb}} \approx 2.68$ Gcal/h, well below the energy ceiling, confirming knee-point identification;
- PPO-Opt achieves H_2S recovery of 99.98% with low variance and fast Aspen convergence (5.5 iterations per episode);
- DDPG-Opt reaches 99.86% recovery with higher convergence variability but slightly lower NH_3 loss;
- DNN surrogates with $R^2 > 0.999$ enable policy transfer with negligible model bias.

The results support the practical feasibility of surrogate-trained RL for industrial process optimization, with a clear path toward dynamic validation and real-plant deployment.

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